CARTESIAN ABELIAN SHEAVES AND CARTESIAN MODULES FORM WEAK SERRE SUBCATEGORIES CLOSED UNDER COLIMITS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-spaces-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the reduction of the modules case to the face maps f_{delta^n_j} relies on the cited lemma-check-cartesian-module (source lines 2706-2710); that lemma's own statement in the frozen source contains an author placeholder (its part 5, 'Add more here.'), recorded when the same label was reviewed as a card candidate - the citation here is to the published Stacks lemma and the present proof is complete; the 5-lemma argument for extensions is at source lines 2695-2700.

Editorial note. The label lemma-Serre-subcat-cartesian-modules names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The simplicial-site situation. Same chapter, Situation \label{situation-simplicial-site}. Statement lines 2669-2670: the simplicial site C_n, f_phi, u_phi with total site C_total.
- The weak Serre subcategory criterion. Homology, Lemma \label{homology-lemma-characterize-weak-serre-subcategory}. Proof lines 2687-2689: the conditions on kernels, cokernels and extensions to be checked.
- Exactness of the simplicial structure functors. Same chapter, \label{lemma-restriction-to-components-site} (card L46149). Proof lines 2691-2694: the restriction functors Sh(C_total) -> Sh(C_n) and the functors f_phi^{-1} are exact, so kernels and cokernels of maps between cartesian sheaves are cartesian.
- Flat pullback exactness and the reduction to face maps. Modules on Sites, Lemma \label{sites-modules-lemma-flat-pullback-exact}; Same chapter, \label{lemma-check-cartesian-module}. Proof lines 2704-2710: the modules case uses the exactness of flat pullback and the fact that it suffices to check the cartesian condition for the maps f_{delta^n_j}.

Source reference. Original lemma 0754: sources/stacks-chow/tags.txt, source lines 9118–9118. Prerequisite pointer invoked at proof line(s) 2690; it names a standard

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fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 07TG: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 2474–2474. Prerequisite pointer invoked at proof line(s) 2709; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 04JC: sources/stacks-chow/tags.txt, source lines 5933–5933. Prerequisite pointer invoked at proof line(s) 2707; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09WE: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 888–888. Prerequisite pointer invoked at proof line(s) 2670; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *In Situation 09WE.*

(1) *The full subcategory of cartesian abelian sheaves forms a weak Serre subcategory of $Ab(\mathcal{C}_{total})$. Colimits of systems of cartesian abelian sheaves are cartesian.*
(2) *Let $\mathcal{O}$ be a sheaf of rings on $\mathcal{C}_{total}$ such that the morphisms*
$$f_{\delta^n_j} : (Sh(\mathcal{C}_n), \mathcal{O}_n) \to (Sh(\mathcal{C}_{n-1}), \mathcal{O}_{n-1})$$
are flat. The full subcategory of cartesian $\mathcal{O}$-modules forms a weak Serre subcategory of $Mod(\mathcal{O})$. Colimits of systems of cartesian $\mathcal{O}$-modules are cartesian.

Proof. To see we obtain a weak Serre subcategory in (1) we check the conditions listed in Homology, Lemma 0754. First, if $\varphi : \mathcal{F} \to \mathcal{G}$ is a map between cartesian abelian sheaves, then $\mathrm{Ker}(\varphi)$ and $\mathrm{Coker}(\varphi)$ are cartesian too because the restriction functors $Sh(\mathcal{C}_{total}) \to Sh(\mathcal{C}_n)$ and the functors f_φ^{-1} are exact. Similarly, if
$$0 \to \mathcal{F} \to \mathcal{H} \to \mathcal{G} \to 0$$
is a short exact sequence of abelian sheaves on $\mathcal{C}_{total}$ with $\mathcal{F}$ and $\mathcal{G}$ cartesian, then it follows that $\mathcal{H}$ is cartesian from the 5-lemma. To see the property of colimits, use that colimits commute with pullback as pullback is a left adjoint. In the case of modules we argue in the same manner, using the exactness of flat pullback (Modules on Sites, Lemma 04JC) and the fact that it suffices to check the condition for $f_{\delta^n_j}$, see Lemma 07TG. □